

Midterm Sample 2: Solutions

ECON 441: Introduction to Mathematical Economics

Instructor: Div Bhagia

1. (5 pts) Answer the following questions.

(a) (1 pt) Consider a mapping $f(x)$. For two distinct values of x , x_0 and x_1 , $f(x_0) = f(x_1)$. Is f a valid function? Answer yes or no. **Yes**

(b) (2 pts) Find the union and intersection for the following sets:

$$A = \{x : x \text{ is an even number}\} \quad B = \{2, 4, 8\}$$

$$A \cup B = A \quad A \cap B = B$$

(c) (1 pt) Consider the following two-variable function:

$$f(x, y) = x + y$$

where $x \in (0, 1)$ and $y \in (0, 1)$. What is the range of f ?

$$(0, 2)$$

(d) (1 pt) Given a system of linear equations $Ax = b$, if $|A| = 5$, what can we say about the solution for this system of equations?

- ☐ Has no solution.
- ☒ Has a unique solution.
- ☐ Has infinitely many solutions.
- ☐ None of the above

2. (5 pts) Consider the following matrix

$$A = I - X(X^T X)^{-1} X^T$$

(a) (3 pts) Is A a square matrix? Show your work or reasoning that led you to this conclusion.

Say the dimension of X is $m \times n$. Then the dimension of $X'_{n \times m} X_{m \times n}$ is $n \times n$. So the dimension of $(X'X)^{-1}$ is also $n \times n$. This implies that the dimension of $X_{m \times n} (X'X)^{-1}_{n \times n} X'_{n \times m}$ is $m \times m$. Hence, $X'X$ and A must be square matrices, but X need not be square.

(b) (2 pts) Prove that A is idempotent i.e. $AA = A$.

$$\begin{aligned} AA &= I - X(X'X)^{-1}X' - X(X'X)^{-1}X' + \underbrace{X(X'X)^{-1}X'X(X'X)^{-1}X'}_I \\ &= I - X(X'X)^{-1}X' = A \end{aligned}$$

3. (8 pts) Consider the following system of equations:

$$x - 2z = 2$$

$$y + z = 12$$

$$x + y + z = 24$$

(a) (1 pt) Write this system of equations in matrix format i.e.,

$$Av = b$$

What is A , v , and b equal to?

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad v = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad b = \begin{bmatrix} 2 \\ 12 \\ 24 \end{bmatrix}$$

(b) (2 pts) Calculate the adjoint of A .

We first need to calculate all the cofactors of A .

$$|C_{11}| = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0 \quad |C_{12}| = -1 \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = 1 \quad |C_{13}| = \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1$$

$$|C_{21}| = -1 \begin{vmatrix} 0 & -2 \\ 1 & 1 \end{vmatrix} = -2 \quad |C_{22}| = \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix} = 3 \quad |C_{23}| = -1 \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = -1$$

$$|C_{31}| = \begin{vmatrix} 0 & -2 \\ 1 & 1 \end{vmatrix} = 2 \quad |C_{32}| = -1 \begin{vmatrix} 1 & -2 \\ 0 & 1 \end{vmatrix} = -1 \quad |C_{33}| = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

$$\text{Adj } A = \begin{bmatrix} 0 & -2 & 2 \\ 1 & 3 & -1 \\ -1 & -1 & 1 \end{bmatrix}$$

(c) (2 pts) Calculate the determinant of A . Is A nonsingular?

$$\begin{aligned} |A| &= a_{11} |c_{11}| + a_{12} |c_{12}| + a_{13} |c_{13}| \\ &= 1.0 + 0.1 + (-2) \cdot (-1) = 2 \end{aligned}$$

A is nonsingular as $|A| \neq 0$.

(d) (1 pt) If you premultiply A^{-1} on both sides of the equation $Av = b$, you should be able to derive an expression to solve for v . Write down this expression.

Premultiplying by A^{-1} :

$$A^{-1}Av = A^{-1}b$$

Since $A^{-1}A = I$, we have $v^* = A^{-1}b$.

(e) (2 pts) Using the expression in (d) solve for v^* .

Since, $A^{-1} = \frac{1}{|A|} \text{Adj} A$

$$\begin{aligned} v^* &= \frac{1}{2} \begin{bmatrix} 0 & -2 & 2 \\ 1 & 3 & -1 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \\ 24 \end{bmatrix}_{3 \times 1} \\ &= \frac{1}{2} \begin{bmatrix} -24 + 48 \\ 2 + 36 - 24 \\ -2 - 12 + 24 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \\ 5 \end{bmatrix} \end{aligned}$$

Checking if it's correct:

$$12 - 2(5) = 2, \quad 7 + 5 = 12, \quad 12 + 7 + 5 = 24$$